Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 16 complete ICs for all displayed variants

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	base10k	0.0356	0.268	1.26	1.01	0.17	0.973	0.0809	0.994	1.13	0	18/18
1 ANN10	soft10k	0.67	1.73	1	46	4.22	3.71	1	0.999	1e+03	0	18/18
1 ANN10	long	0.635	1.28	1	1	2.85	22.5	1	1	15.6	0	18/18
2 P-only DeepONet10	base10k	0.0578	0.114	0.951	0.994	0.165	0.993	0.116	0.995	1.19	0	18/18
2 P-only DeepONet10	soft10k	0.555	1.5	1	1	2.77	1	1	1.35e+04	1	0	18/18
2 P-only DeepONet10	long	0.553	1.54	1	1	3.14	1	1	1.20e+04	1	0	18/18
3 ANN10 + P-only DeepONet10	base10k	0.0373	0.0878	0.901	0.92	0.244	0.993	0.0941	0.99	1	0	18/18
3 ANN10 + P-only DeepONet10	soft10k	0.542	1.81	1	1	5	1	1	0.998	922	0	18/18
3 ANN10 + P-only DeepONet10	long	0.58	1.62	1	0.999	4.07	11	1	0.999	18.5	0	18/18
4 T/P DeepONet10	base10k	0.0436	0.132	1.01	0.779	0.358	0.995	0.104	1.08	1.18	0	17/18
4 T/P DeepONet10	soft10k	0.548	1.69	1	1	4.57	1	1	447	1	0	18/18
4 T/P DeepONet10	long	0.543	1.81	1	1	5.01	1	1	0.98	1.61	0	18/18
5 ANN10 + T/P DeepONet10	base10k	0.0403	0.0677	0.944	0.701	0.148	0.926	0.0543	0.905	0.987	0	18/18
5 ANN10 + T/P DeepONet10	soft10k	0.542	1.81	1	1	5	1	1	0.997	934	0	18/18
5 ANN10 + T/P DeepONet10	long	0.544	1.81	1	1	5.01	1	1	0.999	8.34	0	18/18
6 T/P/ϕ DeepONet10	base10k	0.108	0.0985	0.991	1.02	0.451	1	0.133	1.46	2.34	0	17/18
6 T/P/ϕ DeepONet10	soft10k	0.539	1.81	1	1	5.01	1	1	3.72	1	0	18/18
6 T/P/ϕ DeepONet10	long	0.543	1.81	1	1	5.01	1	1	0.988	3.13	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	base10k	0.0633	0.0986	0.99	0.435	0.195	0.998	0.104	0.957	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	soft10k	0.544	1.81	1	1	4.99	1	1	0.996	830	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	long	0.543	1.81	1	0.999	5.01	1	1	1	3.85	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 88.